# Rajasthan's State Health Insurance and Household Welfare

## A Border-District Difference-in-Differences Study of the Bhamashah Swasthya Bima Yojana (BSBY)

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# Table of Contents


1. Motivation
2. Literature
3. Policy & Question
4. Data & Method
5. Results
6. Cost-effectiveness
7. Conclusions

# Motivation

- Indian households pay a large share of health costs **out of pocket** (OOP); one hospitalisation can push a poor family into debt.
- Policy response: **tax-funded hospital insurance**. Government pays an insurer a premium per family; families get cashless care.
  - National: RSBY (2008) $\rightarrow$ PM-JAY (2018)
  - States: Karnataka, Maharashtra, Tamil Nadu, Gujarat, ...
  - **Rajasthan: BSBY, 13 December 2015**
- Evidence is **mixed**: RSBY did not reduce OOP; Karnataka's scheme cut OOP and mortality.
- **BSBY has not been evaluated against a credible comparison group.**

# Literature (1): Dube, Lester & Reich (2010)

*Review of Economics and Statistics* 92(4)

## Introduction & objective

National studies of US minimum wages found job losses; local case studies did not. Objective: estimate effects using only **counties that touch across a state border**.

## Method

County data 1990–2006 (QCEW); $\sim$316 contiguous cross-border county pairs; county FE + **pair-specific time effects**; SEs clustered by state and border segment.

## Results

Earnings elasticity 0.15–0.23 (all specs). Employment: national specs $-0.18$ to $-0.21$; contiguous pairs $\approx 0$ (0.016). National estimates are biased by **spatial differences in trends**.

**Our use:** the border-pair design, pair fixed effects, clustering.

# Literature (2): Karan, Yip & Mahal (2017)

*Social Science & Medicine* 181

## Introduction & objective

RSBY (2008): Rs 30,000 hospital cover, ∼4 crore families. Earlier studies lacked a comparison group. Objective: causal effect of RSBY on OOP spending.

## Method

NSS consumer expenditure rounds 1999–2000, 2004–05 (pre) and 2011–12 (post), ∼3.5 lakh households. Poorest two quintiles in RSBY vs non-RSBY districts, propensity-score matched. DiD with **two pre-periods** ($\phi_2$ pre-trend, $\phi_3$ effect). States with own schemes dropped.

## Results

No effect on inpatient OOP, its level, or catastrophic spending. Probability of any OOP **rose ∼30%**. Non-medical spending +5%.

**Our use:** comparison group's scheme; OOP benchmark; $\phi_2/\phi_3$; RSBY premium (Rs 560); Rajasthan in early RSBY rollout (p. 84).

# Literature (3): Sood et al. (2014)

*BMJ* 349, g5114

## Introduction & objective

Karnataka's Vajpayee Arogyashree (2010): free tertiary care for below-poverty-line (BPL) households, rolled out first in northern districts. Objective: effect on OOP, hospital use and mortality.

## Method

**Geographic regression discontinuity**: 572 villages on either side of the early/late rollout boundary, propensity-matched; ∼60,000 households surveyed (Sept 2012). Placebos: above-poverty-line households, uncovered conditions.

## Results

Mortality from covered conditions 0.32% vs 0.90% (−64%); OOP for covered tertiary care −64%; utilisation +44% ($p = 0.06$). No effect for ineligible households.

**Our use:** closest in topic and method; placebo logic; village-level RD as future work.

# Literature (4): other studies

| Study | Method / finding | Our use |
|---|---|---|
| Card & Krueger (1994) | NJ vs PA DiD; no job loss | One state vs neighbour |
| Abadie et al. (2010) | Synthetic control; placebo states | Permutation inference |
| Conley & Taber (2011) | Few treated groups: SEs too small | Why permutation |
| Dupas & Jain (2021) | BSBY claims; women under-served | Same scheme, other question |
| Kling et al. (2007) | Standardised effect sizes | SD-unit column |
| Dell (2010) | Spatial regression discontinuity | Future work |
| Dhaliwal et al. (2013) | Cost per unit of causal impact | Cost-effectiveness |
| Finkelstein & McKnight (2008); Hendren & Sprung-Keyser (2020) | Value of risk protection; MVPF | Future welfare work |

**Gap:** no study estimates BSBY's effect with a comparison group.

# The policy: BSBY

**Design** (SIHFW Rajasthan deck)

- Launched 13 Dec 2015, ∼1 crore families
- NFSA + RSBY families, ∼**67% of population**
- Rs 30,000 secondary + Rs 3 lakh tertiary per family per year
- 1,715 packages; cashless; premium paid by state
- Claims paid: **Rs 295.63 cr** (13 Dec 2015 – 10 Feb 2017)

**Timeline**

| | |
|---|---|
| 2005–06 | NFHS-3 (pre-trend) |
| 2008 | RSBY nationally |
| **Dec 2015** | **BSBY launched** |
| Jan–Jul 2016 | NFHS-4 in Rajasthan |
| 2018 | PM-JAY nationally |
| 2019 | BSBY $\rightarrow$ AB-MGRSBY |
| 2019–21 | NFHS-5 (post) |
| 2023–24 | NFHS-6 (third round) |

# Research question and hypotheses

## Question

Did BSBY raise health insurance coverage and reduce out-of-pocket spending, relative to districts just across the state border that had only the national schemes?

- **H1 (coverage):** BSBY raised the share of insured households.
- **H2 (financial protection):** BSBY reduced OOP spending.
- **H3 (use of care):** BSBY changed institutional / public-facility births.
- **Placebos:** no effect on electricity or women's schooling.

**Estimand:** Rajasthan's scheme *relative to* RSBY/PM-JAY, not insurance vs none.

# Data

| Source | Use |
|---|---|
| NFHS-4 (2015–16), NFHS-5 (2019–21) district factsheets | Main analysis (one value per district) |
| NFHS-3/4/5 state factsheets | Pre-trend event study |
| NFHS-6 (2023–24), all border districts, provisional | Third round; cross-checks |
| Census 2011 district map | Finding border pairs |
| SIHFW BSBY presentation | Claims, eligibility (cost) |

**Unit of observation:** district (Census 2011 boundaries), one value per indicator per survey round. Variables and sample: next slide.

# Variables and sample

| NFHS indicator (exact wording) | Unit / treatment | Role | Rounds |
|---|---|---|---|
| Households with any usual member covered under a health insurance/financing scheme | % of households | **Main outcome (H1)** | NFHS-3*, 4, 5, 6 |
| Average out-of-pocket expenditure per delivery in a public health facility | Rs, real 2019–21 (CPI-C); capped at $Q_3+1.5$IQR; also in logs | Welfare outcome (H2) | NFHS-4, 5 (dropped in NFHS-6) |
| Institutional births | % of births in last 5 years | Use of care (H3) | NFHS-4, 5, 6 |
| Institutional births in public facility | % of births in last 5 years | Use of care (H3) | NFHS-4, 5, 6 |
| Population living in households with electricity | % of population | Placebo | NFHS-4, 5, 6 |
| Women with 10 or more years of schooling | % of women 15–49 | Placebo | NFHS-4, 5, 6 |

* state level only (pre-trend test).

## Treatment and time

- $D_i = 1$ if the district is in Rajasthan, 0 if a neighbour
- Pre = NFHS-4 (2015–16); Post = NFHS-5 (2019–21)
- Outcome in the regression: $\Delta Y = Y^{\text{NFHS-5}} - Y^{\text{NFHS-4}}$

## Sample

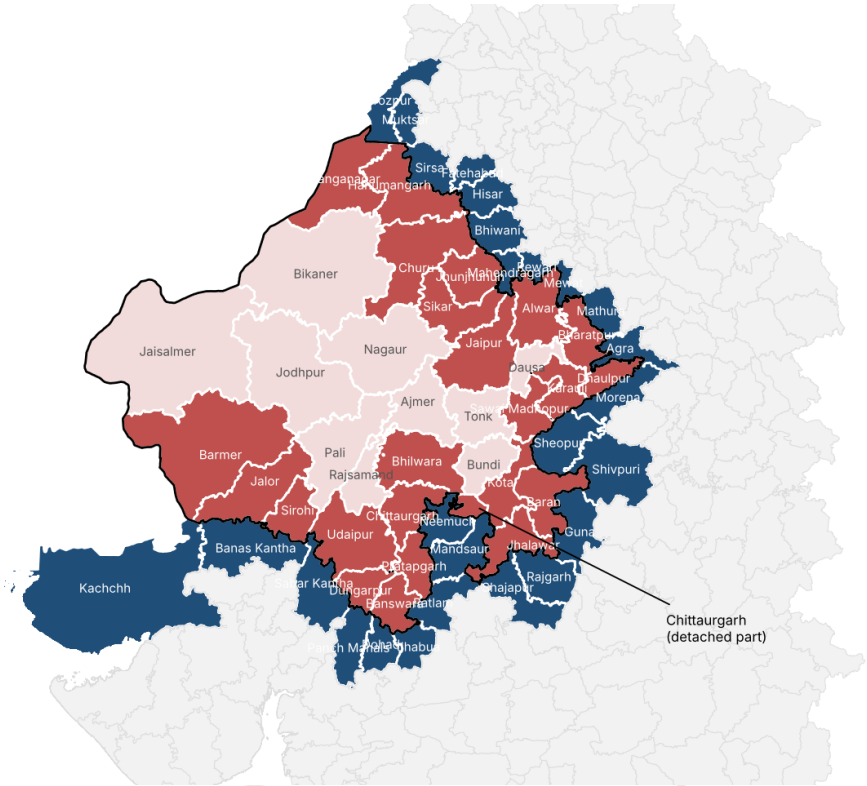
- 23 Rajasthan + 26 neighbour border districts; 51 pairs
- 5 neighbours split after NFHS-4 dropped
- **40 pairs**: 19 Rajasthan + 21 neighbour districts

# Data cleaning (all checks pass)

| Step | Rule / outcome |
|---|---|
| Missing values | No imputation; 0 = suppressed code $\rightarrow$ missing |
| Boundary changes | 5 neighbour districts split after NFHS-4 dropped (51 $\rightarrow$ 40 pairs) |
| Cross-source | 100% exact vs second copy; 33/33 vs NFHS-6 PDF |
| NFHS-6 | Official compendiums, 6 states; 44 border districts found; 220/220 = independent compilation; Gujarat NFHS-5 insurance revision documented |
| Ranges & logic | % in [0,100]; public births $\leq$ institutional births: 0 violations |
| Inflation | OOP to real 2019–21 Rs (CPI-Combined) |
| Outliers | OOP capped at $Q_3 + 1.5\,\mathrm{IQR}$; **no border district reaches the cap** |
| Standardisation | z-scores with NFHS-4 neighbour mean and SD, both rounds |
| Baseline balance | Rajasthan higher OOP (+Rs 594, p = 0.09) $\Rightarrow$ baseline-control spec |

Analysis scripts **refuse to run** unless the audit passes.

# Border sample




**23 Rajasthan × 26 neighbour districts**

- Rajasthan border (treated)
- Neighbour border (control)
- Rajasthan interior (excluded)

- Districts **touching** across the state line (2011 map; ≥ 7 km shared border)
- 51 pairs: MP 22, Haryana 13, Gujarat 10, Punjab 3, UP 3
- **40 pairs** after dropping split districts
- Whole districts only

# Empirical strategy

For district $i$ in contiguous pair $p$:

$$\Delta Y_{ip} = \beta \cdot \mathit{Rajasthan}_i + \gamma_p + \varepsilon_{ip}, \qquad \Delta Y = Y^{\text{NFHS-5}} - Y^{\text{NFHS-4}}$$

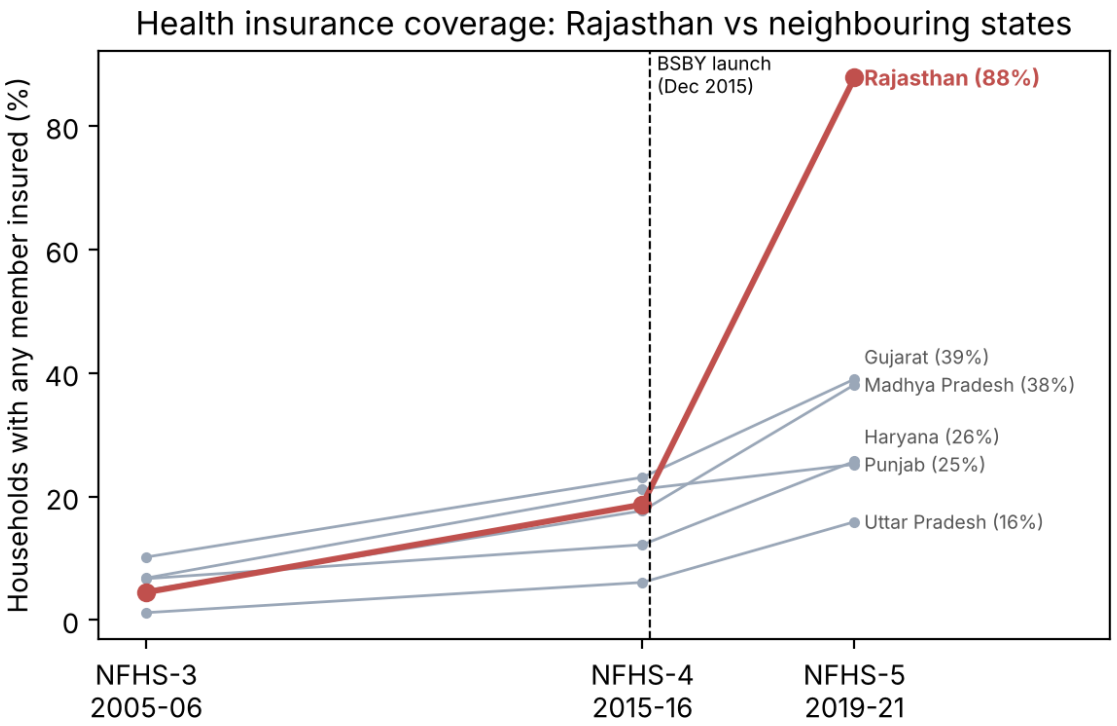
- $\Delta Y$: differencing removes everything fixed about a district (district FE)
- $\gamma_p$: pair fixed effect $\Rightarrow$ compare only with the touching neighbour
- $\beta$: within-pair difference-in-differences
- SEs **two-way clustered** by Rajasthan and neighbour district

## Identifying assumption

**Parallel trends across the border**: without BSBY, both sides would have changed alike. Tested with a three-round state event study.

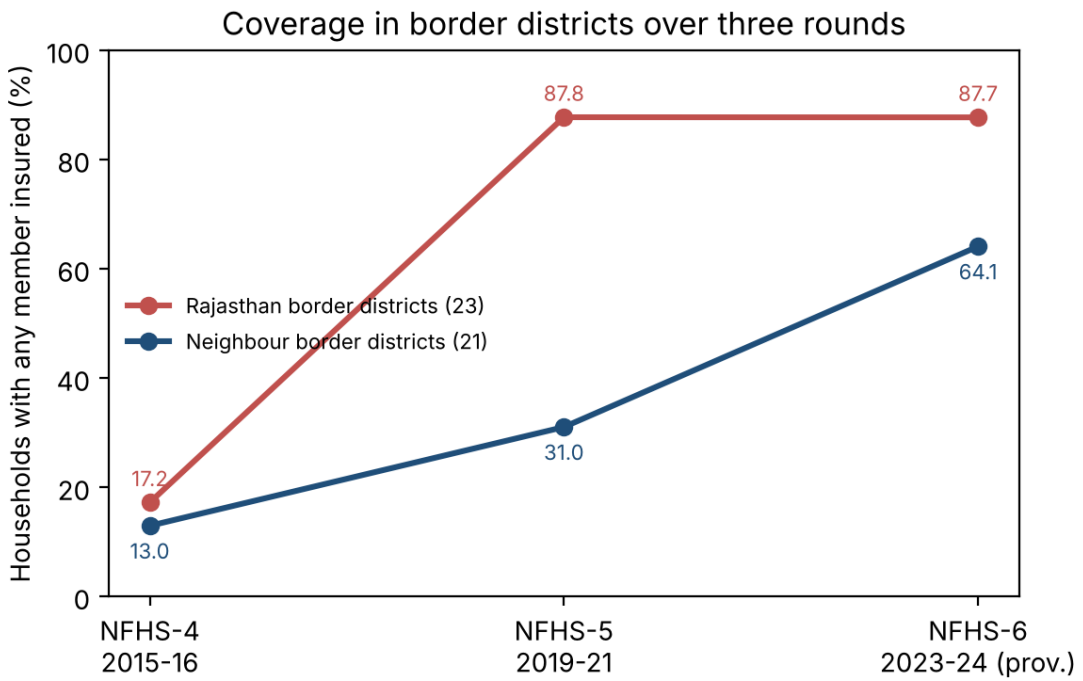
Robustness: drop Gujarat (own scheme); MP border only; pairs $\geq$ 10 km; control for NFHS-4 level; leave-one-out; OOP capping; placebos.

# Pre-trends: state event study




- Pre-period DiD $\phi_2 = +4.1$ pp: roughly parallel
- Post-period DiD $\phi_3 = +56.4$ pp
- Trend-adjusted: **+52.3 pp**
- Permutation: Rajasthan **1st of 27** states, p = 0.037

# Main result: coverage




NFHS-6 is provisional (official IIPS state compendiums). For 3 Gujarat districts the NFHS-6 compendium revises NFHS-5 insurance up ~5 pp.

- Rajasthan: 17.2 $\rightarrow$ 87.8 $\rightarrow$ 87.7%
- Neighbours: 13.0 $\rightarrow$ 31.0%
- Raw DiD: 70.6 $-$ 18.0 = 52.6
- **Regression: $\beta$ = +52.8 pp (SE 5.1), p $<$ 0.001**
- Neighbours reach 64.1% by 2023–24 (next slide)

# Did Rajasthan's lead last?

**Insurance coverage in border districts (% of households)**

| | 2015–16 NFHS-4 | 2019–21 NFHS-5 | 2023–24 NFHS-6 |
|---|---|---|---|
| Rajasthan | 17 | **88** | **88** |
| Neighbours | 13 | 31 | **64** |

- **2015–21:** BSBY puts Rajasthan 53 pp ahead
- **2021–24:** neighbours catch up under PM-JAY
- **2023–24:** Rajasthan still **15 pp** ahead (p = 0.03)

**Takeaway**

BSBY gave Rajasthan near-universal coverage **years before** the national scheme reached its neighbours.

NFHS-6 provisional (official IIPS compendiums). 15 pp = border-pair estimate excluding 3 Gujarat districts with revised NFHS-5 figures. Placebos flat 2019–24; women's schooling +3.6 pp over 2015–24 (p = 0.04).

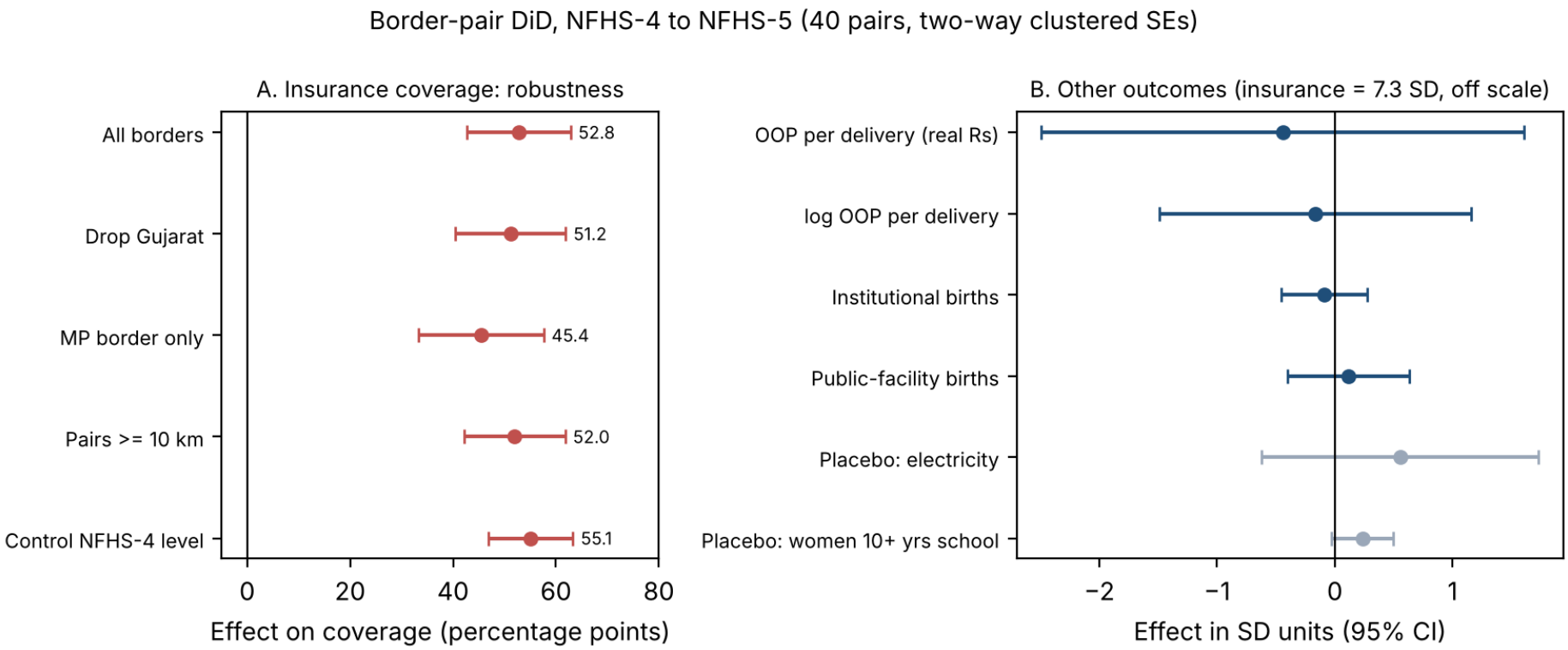
# All outcomes

40 pairs, two-way clustered SEs

| Outcome | $\beta$ (SE) | p | SD units | + NFHS-4 level |
|---|---|---|---|---|
| **Insurance coverage (pp)** | **+52.8 (5.1)** | **<0.001** | 7.3 | +55.1 (4.2) |
| OOP per delivery (real Rs) | −356 (840) | 0.67 | −0.44 | +227 (212) |
| log OOP | −0.08 (0.31) | 0.80 | −0.17 | +0.09 (0.12) |
| Institutional births (pp) | −1.2 (2.3) | 0.62 | −0.09 | +0.8 (1.3) |
| Public-facility births (pp) | +2.4 (5.4) | 0.66 | 0.12 | +5.5 (3.9) |
| *Placebo:* electricity (pp) | +3.2 (3.5) | 0.35 | 0.56 | −0.1 (0.6) |
| *Placebo:* women 10+ yrs school | +3.1 (1.7) | 0.08 | 0.24 | +2.8 (1.5) |

- Coverage: drop Gujarat +51.2; MP only +45.4; pairs $\geq$10 km +52.0; leave-one-district-out +51.3 to +53.9
- Nothing else significant in any sample; women's-schooling placebo borderline

# Robustness at a glance




Panel A: coverage across specifications. Panel B: other outcomes in SD units (95% CI); grey = placebos.

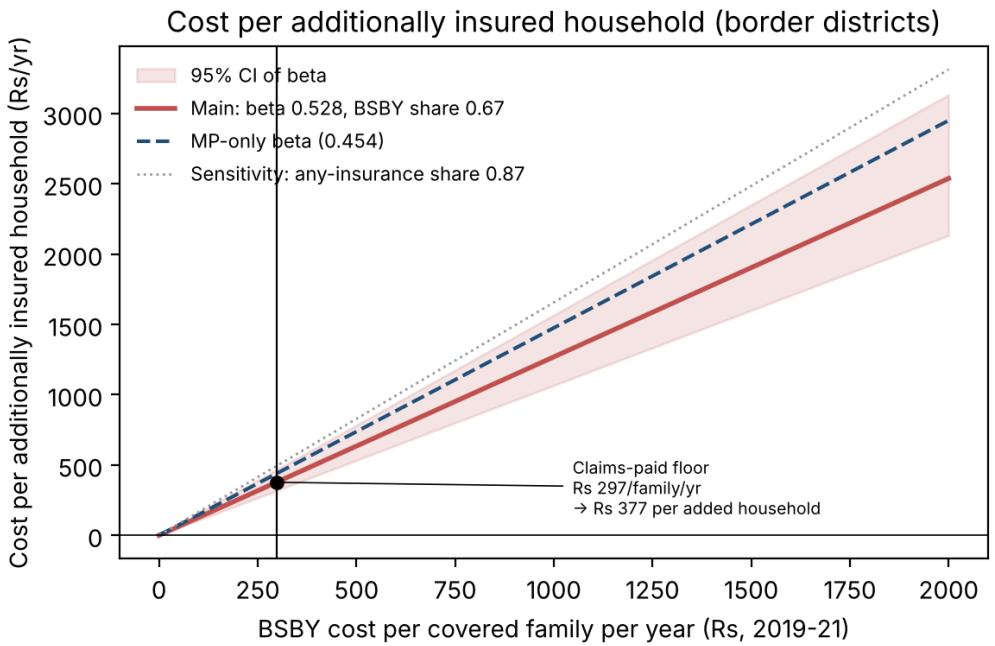
# Out-of-pocket spending

The estimated effect of BSBY on out-of-pocket spending per delivery is −Rs 356 and is not statistically significant. We interpret this as inconclusive rather than as evidence of no effect, for three reasons.

1. **Low precision.** The 95% confidence interval (−Rs 2,003 to +Rs 1,291) is wider than the baseline mean of Rs 2,235. The minimum detectable effect (Rs 2,352) exceeds the baseline, so even a complete elimination of delivery costs would probably go undetected.
2. **Unstable point estimate.** Rajasthan's border districts started with higher spending. Controlling for initial levels reverses the sign (+Rs 227), and excluding a single district (Baran) gives +Rs 30.
3. **Limited relevance of the outcome.** Deliveries in public facilities have been free under JSSK since 2011 in all states. BSBY mainly finances hospitalisations, for which NFHS reports no district-level spending.

These findings are consistent with Karan, Yip & Mahal (2017), who report no reduction in out-of-pocket spending under RSBY.

# Cost-effectiveness




$$\frac{\text{cost/family} \times s_{BSBY} - p_{RSBY}\, c_{cf}}{\beta}$$

- Cost/family = **Rs 297**/yr: claims paid (Rs 295.63 cr, 1.16 yrs, 1 crore families), a *floor* on the premium
- $s_{BSBY} = 0.67$ (SIHFW p. 17)
- $p_{RSBY} = 0$ (early rollout, Karan p. 84)

**Result (incremental cost-effectiveness ratio; Dhaliwal et al. 2013)**

At least **Rs 377** per additionally insured household per year (CI of $\beta$: Rs 317–466; MP-only: Rs 439), measured against the 2019–21 gap. **Break-even:** each such household must gain $\geq$ Rs 377/yr. As neighbours catch up, the cost per unit of *lasting* advantage rises.

# Discussion

- **Coverage:** border estimate (+52.8) and trend-adjusted state estimate (+52.3) agree; by 2023–24 PM-JAY narrowed the gap to $\sim$15 pp, so BSBY's gain was mainly *timing*.
- **Financial protection:** our OOP null agrees with RSBY evidence (Karan et al.) and does not contradict Karnataka's tertiary-care gains (Sood et al.): we measure a margin insurance barely touches, at district resolution.
- **Who benefits:** Dupas & Jain find women use BSBY less; household coverage cannot see this.
- **Value for money:** $\geq$ Rs 377 per additionally insured household; $\sim$40% of spending reaches households that had some insurance anyway.

# Conclusions

1. **BSBY worked as a coverage policy, and worked early:** +53 pp relative to the border by 2019–21, robust to every check; neighbours caught up to within 15 pp by 2023–24.
2. **Financial protection is unproven, not disproven:** OOP per delivery is the wrong margin and the design is underpowered for it.
3. **Cost:** at least Rs 377 per additionally insured household per year (against the 2019–21 gap); benefits must exceed this to justify it.
4. **Policy:** a state scheme can deliver near-universal coverage years before the national scheme does; evaluations should measure hospitalisation spending and health, not just enrolment.

# Limitations and future work

## Limitations

- NFHS-4 baseline began 6 weeks after launch $\Rightarrow \beta$ is a lower bound
- Scheme vs scheme, not insurance vs none
- District averages; large districts
- OOP: deliveries only, underpowered
- One pre-period, state level
- Cost is a claims floor; benefits not monetised
- NFHS-6 provisional; Gujarat NFHS-5 insurance revised
- Long-difference schooling placebo significant (p = 0.04)

## Future work

- NSS 71st vs 75th: OOP per hospitalisation
- NFHS GPS clusters: spatial RD (Dell 2010; Sood et al. 2014)
- Link neighbours' catch-up to PM-JAY rollout
- Actual premium $\Rightarrow$ MVPF / value of risk protection

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# Thank you.

Code, data and full explainer:
`https://github.com/ipsanmoljj/india-welfare-analysis`